

Cell Biology: Structure and Function

Introduction to Cell Biology

The cell is the smallest structural and functional unit of life. Every living organism, from a single bacterium to a human being, is composed of one or more cells. Cell biology studies the physical structure, chemical composition, and functional processes that occur within cells, as well as how cells interact with their environment and with each other.

There are two broad categories of cells: prokaryotic cells, which lack a membrane-bound nucleus (found in bacteria and archaea), and eukaryotic cells, which possess a true nucleus and a variety of specialized internal compartments called organelles. This document focuses on eukaryotic cells, since they are the building blocks of plants, animals, and fungi.

The Cell Membrane

The cell membrane, also called the plasma membrane, is a thin, flexible barrier that surrounds every cell and separates its internal contents from the external environment. It is composed primarily of a phospholipid bilayer: two layers of phospholipid molecules arranged so that their hydrophilic (water-loving) heads face outward and their hydrophobic (water-fearing) tails face inward, forming a stable barrier to most water-soluble substances.

Embedded within the phospholipid bilayer are proteins that perform many critical jobs: channel proteins and carrier proteins regulate what enters and exits the cell, receptor proteins detect chemical signals from outside the cell, and some proteins anchor the cytoskeleton to the membrane. This selective permeability allows the cell to maintain a stable internal environment, a state known as homeostasis, even as conditions outside change.

Mitochondria: The Powerhouse of the Cell

Mitochondria are membrane-bound organelles found in nearly all eukaryotic cells, and they are widely known as the powerhouse of the cell because they generate most of the cell's supply of adenosine triphosphate (ATP), the molecule cells use as their primary energy currency. Mitochondria have two membranes: a smooth outer membrane and a highly folded inner membrane whose folds are called cristae, which greatly increase the surface area available for energy-producing chemical reactions.

The process by which mitochondria produce ATP is called cellular respiration, specifically the stages known as the citric acid cycle (Krebs cycle) and oxidative phosphorylation. During oxidative phosphorylation, mitochondria use oxygen to convert energy stored in glucose and fatty acids into ATP far more efficiently than would be possible without oxygen. Cells with high energy demands, such as muscle cells and neurons, typically contain especially large numbers of mitochondria to meet their metabolic needs.

The Nucleus and DNA

The nucleus is the control center of a eukaryotic cell. It is enclosed by a double membrane called the nuclear envelope, which is perforated by nuclear pores that regulate the movement of molecules, such as RNA and proteins, into and out of the nucleus. Inside the nucleus, genetic material is organized as chromatin, a complex of DNA wound around proteins called histones, which condenses into visible chromosomes when a cell prepares to divide.

DNA (deoxyribonucleic acid) stores the hereditary instructions used for the growth, development, and functioning of the organism. Segments of DNA called genes are transcribed into messenger RNA inside the nucleus; that RNA then travels to the cytoplasm, where it is translated by ribosomes into proteins. The nucleolus, a dense region within the nucleus, is responsible for producing the ribosomal RNA that ribosomes are built from.

Cell Division: Mitosis

Mitosis is the process by which a single eukaryotic cell divides to produce two genetically identical daughter cells. It is essential for growth, tissue repair, and asexual reproduction in eukaryotes. Mitosis is traditionally described in four main phases: prophase, metaphase, anaphase, and telophase, which together ensure that each daughter cell receives a complete and accurate copy of the parent cell's chromosomes.

During prophase, chromatin condenses into visible chromosomes and the mitotic spindle begins to form. In metaphase, chromosomes align along the cell's equatorial plane. In anaphase, sister chromatids are pulled apart toward opposite poles of the cell. Finally, in telophase, two new nuclear envelopes form around the separated chromosomes, and the cell proceeds to cytokinesis, physically splitting into two separate daughter cells.